Certification of Imitation Controllers Using Learned Lyapunov Functions

Master thesis by Josua Christoph Lindemann
Date: July 28, 2026, Date of submission: 16. August 2026

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Darmstadt

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1 Abstract

2 Introduction

3 Optimal Control and Lyapunov Stability Theory

3.1 Discrete-Time Dynamical Systems and Stability

Implementing optimal control schemes on digital hardware requires discrete-time dynamics

$$\dot{x}(t) = f(x(t), u(t)) \tag{3.1}$$

must be discretized, where $x(t) \in \mathcal{X}$ and $u(t) \in \mathcal{U}$ denote the continuous-time state and input vectors, respectively. In this work, the corresponding discrete-time system

$$x_{k+1} = f_d(x_k, u_k) \tag{3.2}$$

is obtained by approximating the continuous dynamics using a numerical integration scheme with a constant sampling time T_s, such as the forward Euler or a fourth-order Runge-Kutta (RK4) method [1].

3.1.1 Lyapunov Stability

Consider the discrete-time closed-loop system under a state-feedback control policy $u = \pi(x)$ with equilibrium $(\mathbf{0}, \mathbf{0})$ such that $f_d(\mathbf{0}, \pi(\mathbf{0})) = \mathbf{0}$. The origin is *locally Lyapunov stable* if there exists a positive definite function $V : \mathcal{X} \to \mathbb{R}_{\geq 0}$ satisfying [2, Theorem 2.13]:

$$\begin{aligned} V(\mathbf{0}) &= 0 \\ V(x) &> 0 \qquad \forall x \in \mathcal{X} \setminus \{\mathbf{0}\} \\ V(f_d(x, \pi(x))) - V(x) &\leq 0 \qquad \forall x \in \mathcal{X} \setminus \{\mathbf{0}\}. \end{aligned} \tag{3.3}$$

If the decrease condition holds strictly along closed-loop trajectories,

$$V(f_d(x, \pi(x))) - V(x) < 0 \qquad \forall x \in \mathcal{X} \setminus \{\mathbf{0}\}, \tag{3.4}$$

the origin is *locally asymptotically stable*, guaranteeing trajectory convergence to the origin for all initial states x_0 in a sufficiently small neighborhood of the origin [2, Theorem 2.13].

3.2 Model Predictive Control and Stability

This section summarizes the Model Predictive Control (MPC) framework and the conditions required to guarantee closed-loop stability in finite-horizon optimal control setups.

3.2.1 Problem Formulation

For the nominal MPC setup considered in this work, the following finite-horizon optimal control problem is solved at each time step k given the current system state x_k [2, Sec. 2.2]:

$$\begin{aligned} \min_{\{x_i\}_{i=0}^{N},\{u_i\}_{i=0}^{N-1}} \quad & \sum_{i=0}^{N-1} \ell(x_i, u_i) + V_f(x_N) \\ \text{s.t.} \quad & x_{i+1} = f_d(x_i, u_i), && i = 0, \dots, N-1, \\ & x_i \in \mathcal{X}, && i = 0, \dots, N, \\ & u_i \in \mathcal{U}, && i = 0, \dots, N-1, \\ & x_0 = x_k. \end{aligned} \tag{3.5}$$

Here, N denotes the prediction horizon, $\ell(x_i, u_i)$ is the stage cost function, and $V_f(x_N)$ represents the terminal cost function. The system dynamics are represented by the discrete-time function $f_d(x_i, u_i)$, while $\mathcal{X}$ and $\mathcal{U}$ denote the state and input constraint sets, respectively.

A quadratic cost function is used for the stage cost, defined as

$$\ell(x_i, u_i) = x_i^\top Q x_i + u_i^\top R u_i, \tag{3.6}$$

where $Q \succ 0$ and $R \succ 0$ are the state and input weighting matrices, respectively [2, Sec. 1.3]. Following the receding horizon principle, only the first control input u_0^* from the optimal sequence is applied to the system, and the optimization is repeated at the next sampling instance [2, Sec. 2.2].

3.2.2 Terminal Ingredients for Stability

To guarantee closed-loop stability in nominal MPC, the optimal control problem Eq. (3.5) is augmented with a terminal constraint $x_N \in \mathcal{X}_f$, where $\mathcal{X}_f \subseteq \mathcal{X}$ denotes the terminal set, and a terminal cost function $V_f(x_N)$ [2, Assumption 2.14]:

$$x_N \in \mathcal{X}_f. \tag{3.7}$$

A set $\mathcal{X}_f$ is positively invariant under a local control law $u = \pi(x_k)$ if any trajectory originating in $\mathcal{X}_f$ remains inside the set for all future time steps [2, Def. 2.9]:

$$f_d(x, \pi(x)) \in \mathcal{X}_f \quad \text{and} \quad \pi(x) \in \mathcal{U}, \qquad \forall x \in \mathcal{X}_f. \tag{3.8}$$

The predictive mechanism is illustrated in Fig. 3.1. While the intermediate states $x_0, \dots, x_{N-1}$ are merely constrained to the safe state space $\mathcal{X}$, the terminal constraint Eq. (3.7) enforces $x_N \in \mathcal{X}_f$. Together with the terminal cost $V_f(x_N)$ acting as a local Lyapunov function, these ingredients guarantee closed-loop asymptotic stability over an infinite horizon [2, Theorem 2.19].

For nonlinear systems, the terminal set $\mathcal{X}_f$ is typically computed using linearized system dynamics around the equilibrium point $(x, u) = (\mathbf{0}, \mathbf{0})$, yielding the linear discrete-time model:

$$x_{k+1} = A x_k + B u_k. \tag{3.9}$$

A widely used choice for these terminal ingredients is then derived from the Linear-Quadratic Regulator (LQR) [2, Eq. (1.18)]. Here, the local control law $\pi(x_k)$ is chosen as a linear state-feedback policy $\pi(x_k) = -Kx_k$, where the optimal LQR gain K is given by

$$K = \left(R + B^\top P B\right)^{-1} B^\top P A. \tag{3.10}$$

Under the standard stabilizability and detectability assumptions, the matrix $P \succ 0$ is obtained as the unique stabilizing solution to the discrete-time Algebraic Riccati Equation (DARE) [2, Eq. (1.18)]:

$$P = A^\top P A - A^\top P B \left(R + B^\top P B\right)^{-1} B^\top P A + Q. \tag{3.11}$$

This matrix parameterizes the quadratic terminal cost function $V_f(x_N) = x_N^\top P x_N$. Accordingly, the terminal set $\mathcal{X}_f$ is defined as an ellipsoidal sub-level set of this cost function [2, Sec. 2.5.5]:

$$\mathcal{X}_f := \left\{ x \in \mathbb{R}^n \;\middle|\; x^\top P x \leq \alpha \right\}. \tag{3.12}$$

Choosing the scaling parameter $\alpha > 0$ sufficiently small guarantees that $\mathcal{X}_f$ satisfies both the state and input constraints $\mathcal{X}$ and $\mathcal{U}$ while ensuring that the quadratic Lyapunov decrease dominates higher-order linearization errors of the nonlinear dynamics.

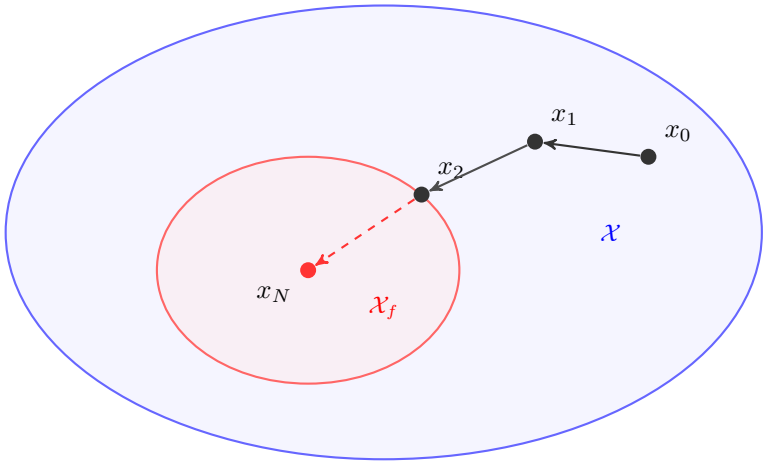

Figure 3.1: Comparison of the terminal set $\mathcal{X}_f$ (red) with the entire state space $\mathcal{X}$ (blue).

3.2.3 Closed-Loop Stability

Under the MPC policy $\pi_{\text{MPC}}(x_k) := u_0^*(x_k)$, the optimal value function $V_N^*(x_k)$ of Eq. (3.5) serves as a Lyapunov candidate over the feasible set $\mathcal{X}_N \subseteq \mathcal{X}$ [2, Sec. 2.4.2]. Using the terminal ingredients from Section 3.2.2, the recursive feasibility at time step $k+1$ is determined by using the remaining inputs of the optimal sequence computed at time step k and appending the terminal control action $\pi(x_N^*)$ [2, Sec. 2.4.2]. Consequently, V_N^* satisfies a decrease condition along closed-loop trajectories, expressed with the relaxed Lyapunov decrease formulation:

$$V_N^*(f_d(x_k, \pi_{\text{MPC}}(x_k))) \leq V_N^*(x_k) - \alpha_N \cdot \ell(x_k, \pi_{\text{MPC}}(x_k)), \qquad \forall x_k \in \mathcal{X}_N, \tag{3.13}$$

where $\alpha_N \in (0, 1]$ denotes the relaxed decre factor [2, Assumption 2.14]. Setting $\alpha_N = 1$ yields the standard nominal Lyapunov decrease

$$V_N^*(x_{k+1}) - V_N^*(x_k) \leq -\ell(x_k, \pi_{\text{MPC}}(x_k)), \tag{3.14}$$

guaranteeing asymptotic stability of the origin **0** for all initial states $x_0 \in \mathcal{X}_N$ [2, Theorem 2.19].

3.3 Stability without Terminal Constraints

only for no terminal constraints -> not asymptotically stable, but can be shown to be practically stable

$$\alpha_N = 1 - \frac{(\gamma - 1)^N}{\gamma^{N-1} - (\gamma - 1)^N} \tag{3.15}$$

if $\alpha_N > 0$ then

$$N < 2 + \frac{\ln(\gamma - 1)}{\ln(\gamma) - \ln(\gamma - 1)}, \tag{3.16}$$

then holds

$$V_N(x_k) \leq \gamma \ell^*(x_k) \tag{3.17}$$

4 Foundations of Neural Control Policies

4.1 Artificial Neural Networks

4.1.1 Multi-Layer Perceptrons

4.1.2 Optimization and Training

4.2 Adversarial Training and Falsification

5 Formal Verification of Neural Control Policies

5.1 Mixed-Integer Linear Programming

5.2 Branch and Bound

5.3 $\alpha\beta$-CROWN

6 Learning Lyapunov-Stable Imitation Controllers

6.1 Neural Policy Approximation and Imitation Learning

This section specifies the data generation process, the neural policy architecture, and the training objective used to approximate the model predictive control (MPC) expert. Specifically, the MPC controller introduced in Section 3.2.1 serves as the expert policy generating the required state-input pairs. The neural policy is subsequently trained to reproduce the expert control actions over the considered state domain.

6.1.1 Data Generation and MPC Expert

The MPC controller introduced in Section 3.2.1 is used as the expert policy and is denoted by π_0. For a given state x, it returns the first input of the optimal MPC solution,

$$\pi_0(x) = u_0^*(x). \tag{6.1}$$

The MPC expert is evaluated along closed-loop trajectories initialized from states sampled within the admissible state space $\mathcal{X}$. At each time step k, the current state x_k and the first control action $u_0^*(x_k)$ returned by the MPC solver are stored as one training pair. If the MPC problem becomes infeasible at any time during one trajectory, all training pairs associated with that trajectory are discarded.

To reduce the overrepresentations of regions with high sampling rates, near-duplicate state-action pairs are removed during preprocessing. The concatenated state-action vectors are normalized and mapped onto an adaptive voxel grid. For each occupied voxel, at most one representative pair is retained.

6.1.2 Policy Network Architecture and Imitation Objective

The policy architecture follows the structure proposed in [3], as it enforces true control inputs at the equilibrium $\pi_\theta(x^*) = u^*$ and satisfies the control input constraints $u_{\min} \leq \pi_\theta(x) \leq u_{\max}$ by construction. The neural policy π_θ is forumlated as

$$\pi_\theta(x) = \text{clamp}\left(f_\theta(x) - f_\theta(x^*) + u^*, u_{\min}, u_{\max}\right). \tag{6.2}$$

where clamp$(\cdot,u_{\min},u_{\max})$ is applied element-wise.

Imitation Objective consists of a state-weighted behavioral cloning loss and a dynamics-aware penalty. The loss terms are normalized using the diagonal scaling matrices

$$S_x = \text{diag}(x_{\text{scale}}) \quad \text{and} \quad S_u = \text{diag}(u_{\text{scale}}), \tag{6.3}$$

where all scaling factors are strictly positive. In addition, the element-wise rectified linear unit (ReLU) is defined as

$$[z]_+ = \max(z,0). \tag{6.4}$$

The normalized distance of the equilibrium is evaluated using the scaled L_1-norm:

$$d(x) = \frac{1}{n_x} \left\| S_x^{-1}(x - x^*) \right\|_1 . \tag{6.5}$$

To assign appropriate priority during behavioral cloning, a state-dependent weight is defined as

$$w_x(x) = w_{\min} + (1 - w_{\min}) \left(\frac{1}{1 + d(x)} \right)^{\alpha}, \tag{6.6}$$

where $w_{\min} \in (0, 1]$ is the baseline weight for states far from the equilibrium and $\alpha \geq 0$ controls the decay rate.

State-Weighted Behavioral Cloning Loss penalizes the normalized Euclidean distance between the neural network policy and the expert targets utilizing the state-dependent weighting. For a batch $\mathcal{X}_\pi$ containing M samples, the behavioral cloning loss is defined as

$$\mathcal{L}_{\text{BC}} = \frac{1}{M n_u} \sum_{x \in \mathcal{X}_\pi} w_x(x) \left\| S_u^{-1} \big(\pi_\theta(x) - \pi_0(x) \big) \right\|_2^2 . \tag{6.7}$$

Dynamics-Aware Constraint Loss applies one-step penalties for successor states that leave the admissible training set $\mathcal{B}_\pi \subseteq \mathcal{X}$. For each state $x \in \mathcal{X}_\pi$, the next state is evaluated under the neural policy as $x_{k+1} = f(x,\pi_\theta(x))$. The loss is defined as

$$\mathcal{L}_f = \frac{1}{M n_x} \sum_{x \in \mathcal{X}_\pi} \left(\left\| \left[x_{k+1} - \bar{x}_\pi \right]_+ \right\|_1 + \left\| \left[\underline{x}_\pi - x_{k+1} \right]_+ \right\|_1 \right). \tag{6.8}$$

Total Imitation Loss is the combination of the state-weighted behavioral cloning and dynamics-aware constraint losses into a single scalar objective. Scaling each term by its respective weight, ω_{BC} and ω_f, yields the overall policy loss:

$$\mathcal{L}_\pi = \omega_{\text{BC}} \mathcal{L}_{\text{BC}} + \omega_f \mathcal{L}_f . \tag{6.9}$$

6.2 Lyapunov Learning for Stability Certification

To establish formal stability guarantees for learned controllers, a neural Lyapunov candidate is trained alongside the policy. A falsification-guided framework is used to find stability violations and therefore make the Lyapunov candidate a valid certificate for stability.

6.2.1 Lyapunov Network Architecture

Following [3], the Lyapunov candidate $V_\theta(x)$ is parameterized to be positive definite by construction:

$$V_\theta(x) = |f_\theta(x) - f_\theta(x^*)| + \left\|(\varepsilon I + R^\top R)(x - x^*)\right\|_1, \tag{6.10}$$

where $f_\theta : \mathbb{R}^n \to \mathbb{R}$ is a neural network, $x \in \mathbb{R}^n$ is the current state, $x^* \in \mathbb{R}^n$ is the equilibrium, $\varepsilon > 0$ is a small regularization constant, and $R \in \mathbb{R}^{n\times n}$ is a learnable matrix. While the first term is only positive semi-definite and may vanish away from the equilibrium, the second term guarantees strict positive definiteness $\forall x \in \mathcal{X} \setminus \{x^*\}$, as the matrix $(\varepsilon I + R^\top R)$ is strictly positive definite by construction for any $\varepsilon > 0$.

6.2.2 Loss Formulation

To ensure that the Lyapunov candidate described in Eq. (6.10) can be certified for stability, the loss function must comprise multiple components. It utilizes the previously introduced ReLU function $[\cdot]_+$ for one-sided penalties, alongside a weighted mean to aggregate the sample losses. Central to the following loss terms is the estimation of the region of attraction (ROA). Throughout the training process, a target level ρ is considered, leading to the corresponding ρ-sublevel set within the training domain $\mathcal{B}_V \subseteq \mathcal{B}_\pi$, defined as

$$\mathcal{V}_\rho := \{x \in \mathcal{B}_V \mid V_\theta(x) \leq \rho\}. \tag{6.11}$$

To prevent this certified region from trivially collapsing towards the origin during optimization, and to ensure numerical stability during normalizations, ρ is strictly constrained by a lower bound $\rho_{\text{min}} > 0$. Furthermore, the next states x^+ needed during loss computation are evaluated on-the-fly using the system dynamics and the policy $u_k = \pi_V(x)$, which may be actively learning during the co-training process.

Relative Lyapunov Decrease Loss penalizes trajectories along which the Lyapunov candidate fails to decay. For each sample, the decrease violation is defined as

$$e_V = (1 - \kappa)V_\theta(x) - V_\theta(x^+), \tag{6.12}$$

where $\kappa \in (0, 1)$ is a hyperparameter that controls the desired rate of decrease. To balance the optimization scale across different magnitudes of V, we normalize this violation to obtain the relative decrease loss:

$$\mathcal{L}_{\text{dec}} = \left[\frac{e_V}{\max(|V_\theta(x)|, \varepsilon_{\text{rel}})}\right]_+, \tag{6.13}$$

where $\varepsilon_{\text{rel}} > 0$ prevents division by zero. Note that the decrease normalization can be omitted depending on the training configuration.

Relative State Invariance Loss penalizes successor states leaving the Lyapunov training domain $\mathcal{B}_V$. To evaluate the relative one-step state-constraint violation, the scaling matrix $S_{\text{range}} = \text{diag}(\bar{x}_V - \underline{x}_V)$ is utilized. The loss per sample for the next state x^+ is computed as

$$\mathcal{L}_{\text{inv}} = \left\|S_{\text{range}}^{-1}\left[x^+ - \bar{x}_V\right]_+\right\|_1 + \left\|S_{\text{range}}^{-1}\left[\underline{x}_V - x^+\right]_+\right\|_1. \tag{6.14}$$

Combined ρ-Gated Condition Loss is the sample-wise ρ-gated combination of the relative Lyapunov decrease loss from Eq. (6.13) and the relative state invariance loss from Eq. (6.14),

$$\mathcal{L}_{\text{cond}} = \mathcal{L}_{\text{dec}} + \omega_{\text{inv}} \cdot \mathcal{L}_{\text{inv}}, \tag{6.15}$$

where ω_{inv} is a hyperparameter balancing the relative importance of the two loss components. However, to apply the ρ-gate appropriately, $\mathcal{L}_{\text{cond}}$ is weighted for each sample using the soft sublevel weight

$$w_\rho = \sigma\left(s \cdot \frac{\rho - V_\theta(x)}{\max(\rho, \varepsilon_{\text{rel}})}\right), \tag{6.16}$$

where $s > 0$ denotes the gate sharpness. This smooth formulation ensures that the weights inside $\mathcal{V}_\rho$ are ≈ 1, whereas the weights for states far outside of ρ approach 0. Evaluating these terms for each sample $x \in \mathcal{X}_\rho$, the weighted mean of the combined condition loss and the ρ-gating weight results in the final loss formulation:

$$\mathcal{L}_{\rho\text{-gated}} = \frac{\sum_{x \in \mathcal{X}_\rho} w_\rho \mathcal{L}_{\text{cond}}}{\sum_{x \in \mathcal{X}_\rho} w_\rho}. \tag{6.17}$$

ROA Loss tries to enlarge the region of attraction by uniformly drawing a batch of M samples $\mathcal{X}_{\text{ROA}} \subseteq \mathcal{B}_V \setminus \xi\mathcal{B}_V$ from the shell of the training space. The loss is then evaluated over these candidates using the following formulation:

$$\mathcal{L}_{\text{ROA}} = \frac{1}{M} \sum_{x \in \mathcal{X}_{\text{ROA}}} \ln\left(\left[\frac{V_\theta(x)}{\rho} - 1\right]_+ + 1\right). \tag{6.18}$$

LIRPA-Condition Loss makes the Lyapunov function more verifier-friendly by including formal violations of the Lyapunov decrease condition as defined in Eq. (6.12), which are identified using the LIRPA framework [4]. In order to compute these violations efficiently, only a filtered subset of regions is evaluated. This subset contains regions for which either the center x_{center}, lower bound x_{lower} or upper bound x_{upper} lies within $\mathcal{V}_\rho$. LIRPA then provides a formal lower bound $e_{V,\text{lower}}$ on the signed Lyapunov decrease $-e_V$ for each of these regions. After evaluation, the LIRPA-condition violation for each region is computed as

$$\mathcal{M}_{\text{L}} = \frac{\ln([-e_{V,\text{lower}}]_+ + 1)}{\max(V_\theta(x_{\text{center}}), \varepsilon_{\text{rel}})}. \tag{6.19}$$

The final LIRPA-condition loss is then computed as the weighted mean over all evaluated regions, where the weight is defined using an exponential function of the squared Euclidean distance between the region center x_{center} and the equilibrium x^*:

$$w_L = \exp\left(-\alpha_{\text{L}} \cdot \|x_{\text{center}} - x^*\|_2^2\right), \tag{6.20}$$

where $\alpha_{\text{L}} > 0$ is a hyperparameter controlling the decay rate of the weight. Finally, the LIRPA-condition loss over N_{L} regions is computed as

$$\mathcal{L}_{\text{L}} = \frac{\sum_{i=1}^{N_{\text{L}}} w_L^{(i)} \cdot \mathcal{M}_{\text{L}}^{(i)}}{\sum_{i=1}^{N_{\text{L}}} w_L^{(i)}}. \tag{6.21}$$

Lyapunov Scaling Loss prevents numerical collapses to 0, by penalizing deviations from the initial average magnitude over a quasi-Monte-Carlo batch $\mathcal{X}_V \subseteq \mathcal{B}_V$ of M samples. The logarithm of the mean Lyapunov value over this batch is then defined as: $\ell_V = \ln\left(\frac{1}{M}\sum_{x\in\mathcal{X}_V} V_\theta(x)\right)$, leading to the scaling loss formulation

$$\mathcal{L}_V = (\ell_{V,0} - \ell_V)^2 . \tag{6.22}$$

Note that the reference value $\ell_{V,0}$ is determined by computing ℓ_V using the initial Lyapunov candidate. Furthermore, to prevent the Lyapunov candidate from overfitting to a fixed set of points, a new batch of samples $\mathcal{X}_V$ is drawn with a specified probability at each optimization step.

Policy Scaling Loss encourages the learned policy π_V to remain close to the output of the original imitation controller π_0, which is based on the MPC teacher. A batch of M samples $\mathcal{X}_\pi \subseteq \mathcal{B}_V$ is generated using a quasi-Monte-Carlo sequence. To prevent overfitting, this batch is redrawn after a specified number of optimization steps. The deviation between the learned policy and the imitation controller is evaluated using a normalized squared difference. The loss is then formulated as

$$\mathcal{L}_\pi = \frac{\sum_{x\in\mathcal{X}_\pi} \|\pi_V(x) - \pi_0(x)\|_2^2}{\sum_{x\in\mathcal{X}_\pi} \|\pi_0(x)\|_2^2} \tag{6.23}$$

R-Matrix Loss regularizes the learnable matrix R by penalizing deviations of its Frobenius norm from its initial value $\ell_{R,0}$:

$$\mathcal{L}_R = (\ell_{R,0} - \|R\|_F)^2 . \tag{6.24}$$

Note that $\ell_{R,0} = \|R\|_F$ is automatically set based on the initial R matrix.

Final Lyapunov Loss represents the combined objective function, summing all previously defined weighted components, given by

$$\mathcal{L}_{\text{total}} = \omega_{\text{cond}}\mathcal{L}_{\rho\text{-gated}} + \omega_{\text{ROA}}\mathcal{L}_{\text{ROA}} + \omega_{\text{L}}\mathcal{L}_{\text{L}} + \omega_V\mathcal{L}_V + \omega_\pi\mathcal{L}_\pi + \omega_R\mathcal{L}_R, \tag{6.25}$$

where the weights ω are hyperparameters that balance the relative importance of the individual loss components.

6.2.3 Falsification-Guided Training

Relying solely on uniform state sampling is not sufficient to reliably detect violations of the Lyapunov condition, especially in higher-dimensional state spaces. To overcome this problem, falsification-guided training actively searches for adversarial counterexamples and incorporates them into the training distribution. This is made possible by an adversarial replay buffer that reconciles the exploration of the estimated attractor region with the targeted exploitation of known error modes.

PGD Falsification is utilized to identify counterexamples that violate the Lyapunov conditions and iteratively inject them into the counterexample pool $\mathcal{C}$. Specifically, an L_∞-Projected Gradient Descent (PGD) is employed to find adversarial states $x_{\text{adv}} \in \mathcal{B}_V$ that maximize the condition violation defined in Eq. (6.15) and Eq. (6.16). The per-sample PGD objective is defined as

$$J_{\text{PGD}} = -w_\rho \mathcal{L}_{\text{cond}}. \tag{6.26}$$

The adversarial search is initialized using the explorative pool $\mathcal{S}$, with up to 25% of the batch supplemented by existing counterexamples to encourage local refinement. The algorithm minimizes J_{PGD} by iteratively updating the input states along the sign of their local gradient with a constant step size α, followed by a projection back onto the training bounds $\mathcal{B}_V$. To retain the strongest violations and mitigate the effects of overshooting, the state yielding the minimum objective value across all K iterations is preserved for each candidate. Ultimately, only states strictly within $\mathcal{V}_\rho$ that exceed the violation tolerance and lie outside the origin exclusion region are retained as counterexamples.

Adversarial Replay Buffer manages the sample distribution for the ρ-gated condition loss $\mathcal{L}_{\rho\text{-gated}}$. It maintains two distinct state pools: an explorative pool $\mathcal{S}$ and a counterexample pool $\mathcal{C}$.

The explorative pool $\mathcal{S}$ is periodically repopulated every outer epoch to balance exploration and local refinement. Every outer epoch, candidate states are oversampled uniformly from the training domain $\mathcal{B}$. The final pool is then constructed using a probabilistic sampling strategy that prioritizes states within an expanded sublevel boundary $m \cdot \rho$, controlled by a scalar margin $m > 1$. As outlined in Algorithm 1, a continuous sampling probability is computed via a sigmoid function, assigning high weights to states inside the estimated region of attraction and exponentially decaying weights to states further outside.

Algorithm 1: Probabilistic Explorative State Pool Resampling

Input: Lyapunov function V_θ, current ROA estimate ρ, margin m, sharpness s, target size N_x
Output: Updated state pool $\mathcal{S}$ of size N_x

1 Generate $4N_x$ candidates: $\mathcal{X}_{\text{ROA}} \sim \mathcal{U}(\mathcal{B}_V)$;

2 For all $x \in \mathcal{X}_{\text{ROA}}$, assign weight $w(x) \leftarrow \sigma\left(s \, \frac{m \cdot \rho - V_\theta(x)}{\max(\rho, 10^{-9})}\right) + \varepsilon$;

3 $\mathcal{S} \leftarrow$ Sample N_x states from $\mathcal{X}_{\text{ROA}}$ with probability $P(x) \propto w(x)$;

The counterexample pool $\mathcal{C}$ stores the critical violations identified during the PGD falsification step. To maintain the relevance of the adversarial states and prevent overfitting to outdated violations, a time-to-live mechanism is implemented. Each injected counterexample is assigned an initial age counter $\tau = 0$, which is incremented upon every subsequent falsification phase. Once a counterexample reaches the maximum age $\tau_{\max}$, it is removed from the buffer. This guarantees that the condition loss continuously penalizes recent violations while allowing the network to gracefully discard resolved edge cases as the Lyapunov candidate evolves.

6.2.4 ρ-Sublevel Estimation

To compute the loss, particularly the ρ-gated condition loss in Eq. (6.17), the target level ρ, which bounds the sublevel set $\mathcal{V}_\rho$, must be continuously estimated during training. Since the goal is to find the

largest level set of the Lyapunov candidate $V_\theta(x)$ that lies entirely within the defined training bounds $\mathcal{B}_V$, a level set can only escape the safe region by intersecting its boundary $\partial\mathcal{B}_V$. Therefore, the maximum certifiable ρ is mathematically strictly determined by the minimum value of the Lyapunov candidate at its boundary. According to the formulation proposed by [3], it is sufficient to evaluate $V_\theta(x)$ exclusively on the boundary $\partial\mathcal{B}_V$ rather than inefficiently sampling the entire state space, in order to estimate the bounds of the ROA. However, since the boundary is a high-dimensional surface and the neural network parameterized in $V_\theta(x)$ forms a highly non-convex landscape, simple uniform sampling on the boundary is insufficient to find the true minimum. Therefore, a Projected Gradient Descent (PGD) approach is employed to actively search for the worst-case boundary points. Starting from uniformly sampled points $x_\partial \sim \mathcal{U}(\partial\mathcal{B}_V)$, PGD iteratively minimizes $V_\theta(x_\partial)$ by taking steps along the negative gradient and projecting the updated states strictly back onto the boundary surfaces. To actively encourage the region of attraction to expand over the course of the co-training process, the formally certifiable limit is intentionally overestimated, as suggested in [3]. Specifically, a target ρ is computed by applying a growth factor $\gamma > 1$ to the boundary minimum found via PGD:

$$\tilde{\rho} = \max\left(\rho_{\min}, \gamma \cdot \min_{x\in\partial\mathcal{B}_V} V_\theta(x)\right). \tag{6.27}$$

By targeting a value for $\tilde{\rho}$ that is slightly above the currently safe maximum, the formulation deliberately leads to small violations near the boundary. These violations are then captured by the aforementioned adversarial replay buffer, effectively forcing the optimization to continuously expand the sublevel set outward. Finally, to ensure numerical stability and prevent irregular jumps in the loss landscape caused by fluctuating local minima across different batches, the ρ value actually used for gating is updated using an exponential moving average (EMA):

$$\rho = \beta\rho + (1-\beta)\tilde{\rho}, \tag{6.28}$$

where $\beta \in [0, 1)$ is the decay rate.

6.3 Formal Verification Methodology

This section adapts the search algorithm for the maximal certifiable sublevel set $\mathcal{V}_\rho$ proposed in [3]. However, iteration limits for scaling and bisection are integrated into Algorithm 2 to reduce computational overhead during evaluation.

To formally certify the Lyapunov candidate, the $\alpha\beta$-CROWN verifier evaluates a logical formulation $\Phi(x;\rho)$ over a bounded certification domain $\mathcal{B}_C := \{x \in \mathbb{R}^n \mid \underline{x}_C \le x \le \bar{x}_C\} \subseteq \mathcal{B}_V$. The goal is to verify that all states in $\mathcal{V}_\rho \cap \mathcal{B}_C$ satisfy the required stability conditions. This implication is rewritten into a disjunctive form that is directly supported by the verifier: a state must either lie outside $\mathcal{V}_\rho$ or satisfy all stability conditions.

The individual logical terms from which this specification is composed are defined as follows:

$$\Phi_{\text{sub}} \equiv V_\theta(x) > \rho, \tag{6.29a}$$
$$\Phi_{\text{pos}} \equiv V_\theta(x) > 0, \tag{6.29b}$$
$$\Phi_{\text{inv}} \equiv \underline{x}_C \le x^+ \le \bar{x}_C, \tag{6.29c}$$
$$\Phi_{\text{dec}} \equiv (1-\kappa)V_\theta(x) - V_\theta(x^+) > 0, \tag{6.29d}$$

Algorithm 2: Certified Value Search via Scaling and Bisection

Input: $\rho_0, \rho_{\min}$, evaluator Φ, scale $s > 1$, tol ε, limits $N_{\text{scale}}, N_{\text{bisect}}$
Output: Max certified value ρ^*

$\rho \leftarrow \max(\rho_0, \rho_{\min})$;

// Phase 1: Scaling

if $\Phi(\rho)$ **then**
  for $k \leftarrow 1$ **to** N_{scale} **do**
    if $\neg\Phi(\rho)$ **then**
      break;
    end
    $\rho_{\text{lo}} \leftarrow \rho$; $\rho \leftarrow \rho \cdot s$;
  end
  $\rho_{\text{up}} \leftarrow \rho$;
else
  for $k \leftarrow 1$ **to** N_{scale} **do**
    if $\Phi(\rho) \vee \rho = \rho_{\min}$ **then**
      break;
    end
    $\rho_{\text{up}} \leftarrow \rho$; $\rho \leftarrow \max(\rho_{\min}, \rho/s)$;
  end
  if $\neg\Phi(\rho)$ **then**
    return 0 ; // No valid value $\geq \rho_{\min}$
  end
  $\rho_{\text{lo}} \leftarrow \rho$;
end

// Phase 2: Bisection

for $k \leftarrow 1$ **to** N_{bisect} **do**
  if $\rho_{up} - \rho_{lo} \leq \varepsilon$ **then**
    break;
  end
  $\rho_{\text{mid}} \leftarrow \frac{1}{2}(\rho_{\text{lo}} + \rho_{\text{up}})$;
  if $\Phi(\rho_{mid})$ **then**
    $\rho_{\text{lo}} \leftarrow \rho_{\text{mid}}$;
  else
    $\rho_{\text{up}} \leftarrow \rho_{\text{mid}}$;
  end
end

return ρ_{lo};

where $x^+ = f(x, \pi_\theta(x))$ denotes the discrete one-step state transition. Here, Eq. (6.29a) determines whether x lies outside the target sublevel set $\mathcal{V}_\rho$. For states within $\mathcal{V}_\rho$, Eq. (6.29b) enforces strict positive definiteness away from x^*, Eq. (6.29c) guarantees forward invariance within $\mathcal{B}_C$, and Eq. (6.29d) ensures the strict Lyapunov decrease property.

Finally, combining the sublevel exclusion with the joint stability conditions yields the total specification evaluated by the verifier:

$$\Phi(x;\rho) \equiv \Phi_{\text{sub}} \vee \left(\Phi_{\text{pos}} \wedge \Phi_{\text{inv}} \wedge \Phi_{\text{dec}}\right). \tag{6.30}$$

The equilibrium x^* must be excluded from the certification domain $\mathcal{B}_C$. By construction, $V_\theta(x^*) = 0$ and $x^+ = x^*$ hold at the equilibrium, causing the strict positivity (Φ_{pos}) and strict decrease (Φ_{dec}) conditions to evaluate to false at x^*. Since these conditions hold only for $x \neq x^*$ and positive definiteness is enforced by construction, formal verification is performed over the punctured domain $\mathcal{B}_C \setminus \mathcal{B}_{\text{hole}}$. Here, $\mathcal{B}_{\text{hole}}$ denotes a small hyper-rectangular region centered at x^*. The punctured certification domain is thus split into multiple subregions, as shown in Fig. 6.1.

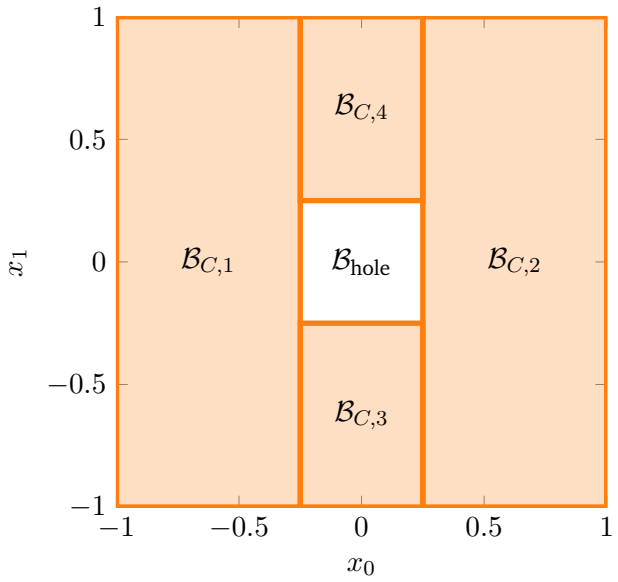

Figure 6.1: 2D Hole exclusion example with $\mathcal{B}_{\text{hole}} := \left\{x \in \mathbb{R}^2 \;\middle|\; \|x\|_\infty < 0.25\right\}$.

7 Experimental Evaluation and Certification Results

7.1 System Modeling and Problem Formulation

For our experimental setup, two benchmark systems are selected: the double integrator and the cartpole. These systems provide a good balance between simplicity and complexity allowing us to certify the stability of the closed-loop system.

7.1.1 Double Integrator

The double integrator is a simple second-order linear system; therefore, it serves as the ideal baseline system in the learning and certification pipeline. Discretizing the continuous-time dynamics using the explicit Euler method with a sampling time of $\Delta t = 0.1$, leads to the discrete-time state-space equation:

$$x_{k+1} = f_{\text{DI},d}(x_k, u_k) = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ \Delta t \end{bmatrix} u_k \tag{7.1}$$

where $x_k = \begin{bmatrix} p_k & v_k \end{bmatrix}^\top$ is the state vector containing the position and velocity of the system, respectively, and u_k denotes acceleration.

The *MPC setup* is formulated according to the problem in Eq. (3.5) subject to the double-integrator dynamics in Eq. (7.1). A prediction horizon of $N = 20$ steps is chosen, and the states and inputs are penalized using the weighting matrices $Q = \text{diag}(15, 1)$ and $R = 0.1$.

7.1.2 Cartpole

The inverted pendulum on a cart, commonly referred to as the cartpole system, is used as a nonlinear benchmark system in this work. There are four states in the system: the cart position p, the cart velocity v, the pendulum angle θ, and the pendulum angular velocity ω. Here, the state vector is defined as $x = \begin{bmatrix} p & v & \theta & \omega \end{bmatrix}^\top$, and the input u is the force F applied to the cart. This leads to the following continuous-time dynamics of the cartpole system:

$$f_{\text{CP}}(x,u) = \begin{bmatrix} \dot{p} \\ \dot{v} \\ \dot{\theta} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} v \\ \frac{F+ml\omega^2\sin(\theta)-mg\sin(\theta)\cos(\theta)}{M+m-m\cos^2(\theta)} \\ \omega \\ \frac{-F\cos(\theta)-ml\omega^2\sin(\theta)\cos(\theta)+(M+m)g\sin(\theta)}{l(M+m-m\cos^2(\theta))} \end{bmatrix} \tag{7.2}$$

where $M = 1\,\text{kg}$ is the mass of the cart, $m = 0.1\,\text{kg}$ is the mass of the pendulum, $l = 0.8\,\text{m}$ is the length of the pendulum, and $g = 9.81\,\text{m/s}^2$ is the acceleration due to gravity. Discretizing the continuous-time dynamics using the explicit Euler method with a sampling time of $\Delta t = 0.05\,\text{s}$, leads to the discrete-time state-space equation:

$$x_{k+1} = f_{\text{CP},d}(x_k, u_k) = x_k + \Delta t f_{\text{CP}}(x_k, u_k) \tag{7.3}$$

For the nonlinear benchmark, a prediction horizon of $N = 40$ stages is chosen. The system states and control inputs are penalized within the optimization framework of Eq. (3.5) using the weighting matrices $Q = \text{diag}(10, 1, 10, 0.01)$ and $R = 0.5$, respectively, subject to the dynamics defined in Eq. (7.3).

7.2 Linear System Results: Double Integrator

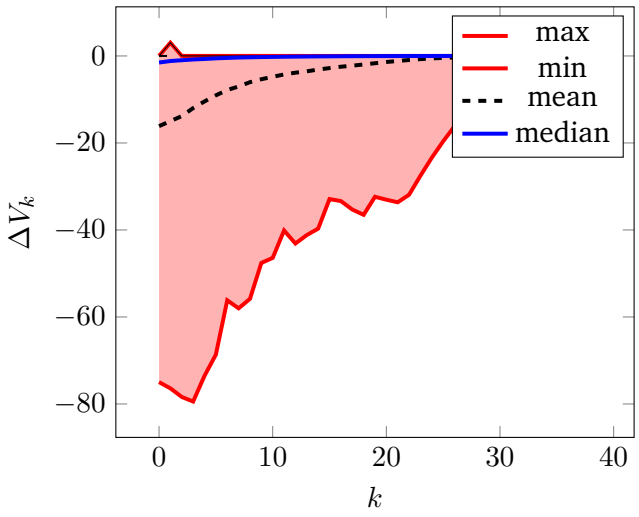

Figure 7.1: Double integrator cost descent. $\Delta V = V_N(x_{k+1}) - V_N(x_k)$

7.3 Nonlinear System Results: Cartpole

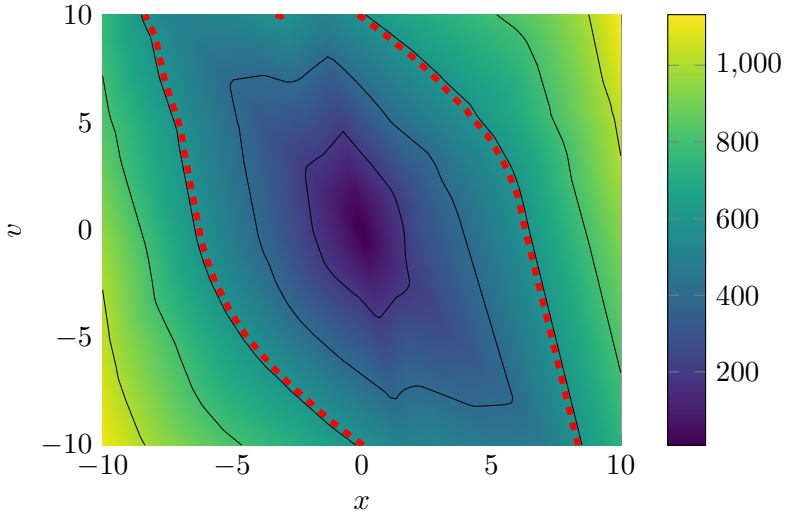

Figure 7.2: Double integrator Lyapunov landscape and ROA: value

8 Discussion

8.1 Qualitative Discussion of Results

8.2 Challenges with Nonlinear Dynamics

8.3 Scalability and Limitations

9 Conclusion

A Appendix

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